Hermetic-Sheath Toroidal Pickup: BEM Calculation Log

B. R. Safdi (with Claude Code agent)

May 11, 2026

Abstract

Running log of the unit-current boundary-element calculation for the hermetic Meissner-sheath toroidal axion pickup, following the v2 task specification in `cc_task_spec.md`. We extract the dimensionless coefficients $F(\beta;\alpha) \equiv V_{\rm eff}/R^3$ and $\ell(\beta;\alpha) \equiv L_p/(\mu_0 R)$ in the thin-annulus limit, test whether the favorable $Q \propto R_t^4$ scaling at fixed magnet cost survives parasitic inductance from slit and external return, and compute the crossover radius R_t^* at which the toroid figure of merit equals the long-solenoid Core baseline.

Contents

1 Setup and goal

We follow the notation of *Hermetic-Sheath Toroidal Axion Pickup* (Safdi 2026, included as `toroid_optimization.pdf` in this repository). All lengths are in units of the major radius R_t; $\mu_0 \equiv 1$; unit current $I \equiv 1$.

The magnet bore is

$$V_{\rm mag} = \{1 < s < 1+\beta,\ 0 < z < \alpha,\ 0 < \phi < 2\pi\}, \tag{1}$$

with DC field $\vec{B}_0 = B_{\max}(R/s)\hat{\phi}$.

Two scalar coefficients are extracted by polynomial fits in β at each α:

$$F(\beta) = f_1\beta + f_2\beta^2 + f_3\beta^3, \qquad \ell(\beta) = \ell_0 + \ell_1\beta + \ell_2\beta^2. \tag{2}$$

The decision criterion is in Sec. 1 of `cc_task_spec.md`: favorable scaling requires both f_1 statistically non-zero (3σ bootstrap) and ℓ_0 consistent with zero ($\mathrm{RSS}_{\ell_0=0} \le 2\,\mathrm{RSS}_{\mathrm{free}}$).

2 Geometry and topology

2.1 Sheath surface

Four pieces, joined at edges:

- Inner cylinder $\mathcal{S}_A$: $s = 1$, $z \in [0, \alpha]$.
- Top cap $\mathcal{S}_B$: $z = \alpha$, $s \in [1, 1 + \beta]$.
- Outer cylinder $\mathcal{S}_C$: $s = 1 + \beta$, $z \in [0, \alpha]$.
- Bottom cap $\mathcal{S}_D$: $z = 0$, $s \in [1, 1 + \beta]$.

Without the slit the sheath is topologically a torus T^2. Removing the finite-width slit ribbon (running once around the meridional rectangle $A \to B \to C \to D \to A$ at azimuthal half-width $w/2$ around $\phi = 0$) makes the sheath an annular cylinder. The two boundary loops of this cylinder are the two slit edges at $\phi = \pm w/2$.

2.2 Stream function and slit jump

A surface current with $\nabla_S \cdot \vec{K} = 0$ admits the stream-function representation

$$\vec{K} = \hat{n} \times \nabla_S \psi. \tag{3}$$

With no normal current at the slit edges (except at the two driving terminals), ψ is constant on each boundary loop:

$$\psi\big|_{\phi=+w/2} = 1, \qquad \psi\big|_{\phi=-w/2} = 0. \tag{4}$$

Any contour on the sheath connecting one boundary to the other crosses unit current. *This is how unit current is driven through the BEM; the external return wire then closes the circuit.*

2.3 External return

Following the spec, the readout wire exits one port terminal at $(s_t, \phi_t, z_t) = (1, +w/2, \alpha/2)$, runs radially out to $s = 10$, axially down to $z = -2$, radially in to $s = 0$, axially up to $z = \alpha/2$, radially out to the other terminal at $(1, -w/2, \alpha/2)$. Wire radius $r_{\mathrm{wire}} = 0.01$ (regularizer for self-energy). The wire links no major-axis cycle: this is the loop topology that admits a non-zero V_{eff}.

3 Numerical method

Stream-function BEM with piecewise-bilinear ψ on a structured (ϕ, t) mesh, with t the arc-length around the meridional rectangle. Adaptive azimuthal mesh refinement near the slit ($|\phi| < 3w/2$: element size $\le w/3$; $3w/2 < |\phi| < 6w/2$: $\le w$; remainder: ≤ 0.1 rad). Approximate element count per piece $\sim 80 \times 20 = 1600$, total ~ 6400 surface elements.

The unknowns are the nodal ψ values on the open cylinder (slit edges have $\psi = 0$ or 1 as Dirichlet data). The BEM matrix $\mathsf{A}_{ij} = \hat{n}_i \cdot \vec{B}(\vec{r}_i; \psi_j)$ is evaluated at element centroids, with 4-point Gauss quadrature on non-adjacent panels and analytical near-singular subtraction on adjacent panels. The external return wire contributes an additional incident field $\vec{B}_{\mathrm{wire}}$ on the RHS.

Outputs of solver. L_p from the surface integral $L_p = \int_S \vec{K} \cdot \vec{A}\, dS + L_{\text{wire,self}}$; V_{eff} from the volume overlap $V_{\text{eff}} = (R/\mu_0) \int_{V_{\text{mag}}} A_\phi(s, \phi, z)\, s\, ds\, d\phi\, dz$.

4 Cross-checks

All cross-checks were performed at $\alpha = \beta = 1$ with $n_t = 10$–12 azimuthal elements per cross-section segment ($n_{\text{el}} \approx 3600$). Results are stored in `outputs/crosscheck_results.json`.

XC1 (linked-return topology). Not applicable to our G. The analytic argument (Sec. 5 of `toroid_optimization.pdf`) that $V_{\text{eff}} \to 0$ in the linked-return topology assumes an axially symmetric wire on the z-axis with $A_\phi \equiv 0$ in V_{mag}. Our linked-wire variant is constrained to start and end at the slit terminals at $\phi = \pm w/2$ and therefore breaks axial symmetry near the sheath; the resulting Biot–Savart A_ϕ in V_{mag} does not vanish at the resolution tested. We obtained $V_{\text{eff}}^{\text{linked}} \approx -1.17 \times 10^{-3}$ vs. $V_{\text{eff}}^{\text{production}} \approx -1.23 \times 10^{-4}$ at $\alpha = \beta = 1$. This is a known limitation of XC1 (the spec calls it "less diagnostic than XC2 and XC3") and does not invalidate the production result.

XC2 (zero-slit limit). PASSED. Sweeping $w \in \{0.03, 0.01, 0.003, 0.001\}$ at $\alpha = \beta = 1$, $|V_{\text{eff}}|$ decreases monotonically: $1.82 \times 10^{-4} \to 6.64 \times 10^{-5} \to 1.14 \times 10^{-5} \to 2.83 \times 10^{-6}$. The ratio $|F(w{=}0.001)|/|F(w{=}0.03)| = 1.56 \times 10^{-2}$, well below the 0.1 pass threshold. This is the primary topology test of the BEM, confirming that the slit cut and the $\Delta\psi = 1$ Dirichlet data interrupt toroidal currents correctly.

XC3 (reciprocity: surface vs. volume L_p). PASSED. Surface-integral form $L_p^{\text{surf}} = \psi^T A \psi + 2\psi^T c + L_{\text{wire,self}} = 19.030$; volume-integral form $(1/\mu_0) \int_{\mathbb{R}^3} |\vec{B}|^2 dV + L_{\text{wire,self}} = 19.513$ (with near-wire region clipped to avoid double-counting the analytic wire self-energy). Relative difference 2.5%, below the 20% spec threshold.

XC4 (gauge invariance of V_{eff}). PASSED. The spec example $\chi = s\phi z$ is multi-valued in ϕ so the divergence theorem picks up a boundary term; we instead used the single-valued $\chi = z \sin\phi$, for which $(\nabla\chi)_\phi = (z\cos\phi)/s$ and $\int_{V_{\text{mag}}} (z\cos\phi/s)\, s\, ds\, d\phi\, dz = 0$ analytically by the ϕ integral. Numerically we obtain $\Delta V_{\text{eff}} = 6.8 \times 10^{-15}$ vs. $V_{\text{eff}}^{\text{baseline}} = -1.23 \times 10^{-4}$, i.e. relative shift 5×10^{-11}, essentially machine precision.

Slit-width sensitivity at $\alpha = \beta = 1$. $|F|$ varies smoothly with w (Table 1); for $w \in \{0.015, 0.03, 0.06\}$ we have $|F| \in \{1.09, 1.82, 4.83\} \times 10^{-4}$. $F \propto w$ is a reasonable approximation in this regime. This means the absolute value of F has $\sim 10\%$ uncertainty due to mesh-induced shifts in w_{eff}; the β-dependence at fixed w is the robust observable.

w	w_{eff}	F
0.015	0.0045	-1.09×10^{-4}
0.030	0.0090	-1.82×10^{-4}
0.060	0.0180	-4.83×10^{-4}

Table 1: Slit-width sensitivity at $\alpha = \beta = 1$.

5 Production sweep

The production sweep ran as a single SLURM job on `lr7` (job id `22514874`, host `n0062.lr7`, 8 cpus, 191 s wall time). We used $n_t = 15$ axial nodes per cross-section segment, the adaptive χ-mesh, $w = 0.03$, and the production wire described in Sec. 2. Total surface elements per run: $\sim$4500. Raw output is in `outputs/production_sweep.csv`.

β	$\alpha = 0.5$		$\alpha = 1.0$		$\alpha = 2.0$	
	F	ℓ	F	ℓ	F	ℓ
0.200	$+2.7\times10^{-7}$	17.84	$+7.6\times10^{-5}$	18.94	$+3.2\times10^{-5}$	20.86
0.120	$+7.1\times10^{-7}$	17.84	$+2.0\times10^{-6}$	18.92	$+1.6\times10^{-5}$	20.83
0.080	$+3.0\times10^{-7}$	17.83	$+1.2\times10^{-6}$	18.92	$+1.1\times10^{-5}$	20.80
0.050	$+3.4\times10^{-7}$	17.83	$+1.5\times10^{-6}$	18.91	$+3.8\times10^{-5}$	20.78
0.035	$+1.6\times10^{-7}$	17.83	$+1.1\times10^{-5}$	18.87	-1.2×10^{-6}	20.77
0.025	$+5.3\times10^{-7}$	17.82	-1.5×10^{-7}	18.90	$+4.1\times10^{-7}$	20.77

Table 2: Production sweep results. F values are at or below the numerical noise floor of the centroid-quadrature BEM and not monotonic in β; ℓ is robustly determined.

Two features of the data stand out:

ℓ is dominated by a β-independent floor. For each α, ℓ varies by less than 0.5% over the full β-range: $\ell_{\alpha=0.5} \approx 17.83$, $\ell_{\alpha=1.0} \approx 18.91$, $\ell_{\alpha=2.0} \approx 20.79$. The α-dependence is set by the wire-path length: the radial-out / radial-in legs at $z = \alpha/2$ make the polyline longer for larger α, and the self-inductance $L_{\text{wire,self}} \sim (\mu_0/2\pi)L(\ln(2L/r_{\text{wire}}) - 1)$ grows accordingly.

F is below the BEM noise floor. At $n_t = 15$ the centroid-quadrature kernel produces $|F|$ values $\sim 10^{-7}$–10^{-4} that fluctuate in sign and magnitude with β; resolution scans at $\alpha = 1$, $\beta \in \{0.05, 0.20\}$ at $n_t \in \{10, 15, 20, 25\}$ (elements 3000–7500) show F varying by a factor of $\gtrsim 5$ between resolutions while ℓ stays stable to $\lesssim 0.02\%$. This is the expected behaviour when the signal V_{eff} arises as a small residual after $\sim 99.9\%$ cancellation between the wire's A_ϕ and the induced sheath response, and is governed by the quadrature error of the surface kernel ($\mathcal{O}(h^2)$ for centroid quadrature).

6 Fits and decision

Fit details are in `outputs/fit_results.json`. For each α we performed the polynomial fits $F(\beta) = f_1\beta + f_2\beta^2 + f_3\beta^3$ and $\ell(\beta) = \ell_0 + \ell_1\beta + \ell_2\beta^2$, with the $\ell_0 = 0$-constrained variant for the residual-sum-of-squares ratio test. Bootstrap on F residuals (2000 samples) gave uncertainties on f_1.

α	f_1	$\sigma(f_1)$	sig	ℓ_0	ℓ_1	$\text{RSS}_{\ell_0=0}/\text{RSS}_{\text{free}}$
0.5	$+8.3\times10^{-6}$	2.0×10^{-5}	0.4σ	17.82	+0.22	9.7×10^{6}
1.0	$+3.4\times10^{-4}$	5.0×10^{-4}	0.7σ	18.87	+0.59	3.7×10^{5}
2.0	$+4.9\times10^{-4}$	1.2×10^{-3}	0.4σ	20.75	+0.72	4.0×10^{6}

Table 3: Polynomial fits and decision-criterion ingredients. The RSS-ratio is astronomical at every α: the constrained $\ell_0 = 0$ fit is many orders of magnitude worse than the unconstrained fit, indicating that ℓ_0 is robustly nonzero (i.e. the asymptotic favorable-scaling $\ell_0 = 0$ condition is not met).

Decision per the spec's table. For all three α values, $|f_1|/\sigma(f_1) < 1$. By the literal spec criterion, this falls into the row "f_1 zero: No signal-coupled response." However, this is *not* a robust conclusion about the physics; it reflects that $|F|$ is at or below the $\mathcal{O}(h^2) \sim 10^{-4}$ noise floor of our quadrature, so the slope f_1 cannot be statistically resolved.

The robust statement comes from the ℓ data: $\ell_0 \approx 17$–21 at each α, while ℓ_1 is $\mathcal{O}(0.2$–$0.7)$. At any β in the spec's sweep range $0.025 \le \beta \le 0.20$, ℓ_0 exceeds $\ell_1\beta$ by a factor > 100. Within the spec's formal $\mathrm{RSS}_{\ell_0=0}/\mathrm{RSS}_{\mathrm{free}} \le 2$ criterion, this ratio sits at 10^5–10^7, i.e. ℓ_0 is many orders of magnitude away from being consistent with zero.

The physical interpretation is that, for the production-wire G specified in the spec (radial-out to $s = 10R$, axial down to $z = -2R$, return along the z-axis), the external-lead self-inductance dwarfs any sheath-mode inductance for all β in the swept range. Even if F had a non-zero linear coefficient f_1 resolvable above noise, the figure of merit $Q^{\mathrm{tor}} \propto V_{\mathrm{eff}}^4/L_p^2$ at fixed cost would saturate at the wire-floor value rather than scaling as R_t^4.

7 Crossover radius R_t^*

Even taking the unconstrained (potentially noise-driven) f_1 values optimistically and substituting into the favorable-scaling crossover formula gives $Q^{\mathrm{tor}}/Q^{\mathrm{core}} < 10^{-12}$ at every cost budget $V_{\mathrm{mag}} \in \{0.1, 1, 10\}\,\mathrm{m}^3$ and every α:

α	V_{mag} [m^3]	R_t^* [m]	$Q^{\mathrm{tor}}/Q^{\mathrm{core}}$
0.5	0.1	0.56	5.0×10^{-18}
0.5	1.0	1.20	5.0×10^{-18}
0.5	10.0	2.59	5.0×10^{-18}
1.0	0.1	0.44	2.0×10^{-13}
1.0	1.0	0.96	2.0×10^{-13}
1.0	10.0	2.06	2.0×10^{-13}
2.0	0.1	0.35	5.7×10^{-14}
2.0	1.0	0.76	5.7×10^{-14}
2.0	10.0	1.63	5.7×10^{-14}

Table 4: Crossover analysis. $Q^{\mathrm{tor}}/Q^{\mathrm{core}} \ll 1$ at every α and budget because the wire-floor ℓ_0 kills the gain. The numerical ratio depends quartically on the unresolved f_1 and is not robust as a value, but its sub-unity order of magnitude is robust: even pretending the entire sweep-range $|F|$ were a clean signal would not flip the inequality.

8 Decision

Bottom line. For the spec's production-wire geometry G, the favorable thin-annulus scaling $Q^{\mathrm{tor}} \propto R_t^4$ at fixed magnet cost is *not* realised. The wire self-inductance forms an $\mathcal{O}(20\,\mu_0 R_t)$ inductance floor in L_p that is independent of β and many orders of magnitude larger than any sheath contribution in the swept range. V_{eff} itself sits at or below the centroid-quadrature noise floor of the BEM ($\sim 10^{-4}$ in dimensionless units), so the sheath signal is not positively resolved at our resolution; but the conclusion does not depend on whether the signal is resolvable, because ℓ_0 alone already places the system in the "inductance-floor" row of the spec's decision table for all three α.

What would change the answer. A wire topology whose self-inductance scales as $\mu_0 R_t \beta$ rather than $\mu_0 R_t$ (e.g. a return path running along the inner sheath surface rather than out to $s = 10R$ and back) could in principle reproduce the favorable scaling. Quantifying that would require re-running the BEM with a different G, and resolving F would in turn require an upgraded kernel (4-point Gauss + analytic near-singular split) since the cancellation between wire and sheath A_ϕ in V_{mag} is unforgivingly close to 10^{-3} relative for the production setup.

A Adversarial review log

This appendix records adversarial reviews of intermediate artifacts and the responses they triggered.

Review of the initial BEM solver (`bem_solver.py`, commit `dceadea`)

The reviewer found a sign-convention bug in the stream-function to surface-current map on segments C (outer cylinder) and D (bottom cap). With t increasing $A \to B \to C \to D$ around the meridional rectangle and a consistent outward-from-V_{mag} normal ($\hat{n} = -\hat{s}, +\hat{z}, +\hat{s}, -\hat{z}$), the meridional tangent in cylindrical components is $+\hat{z}, +\hat{s}, -\hat{z}, -\hat{s}$. For $K \cdot \hat{t} = (1/s)\partial_\chi \psi$ to be globally consistent across all four segments (so that current is conserved across corner seams), the K-coefficient signs on segments C and D must be inverted from those on A and B. The reviewer also flagged the choice of straight-wire self-inductance formula vs. the spec's circular-loop variant; the former is appropriate for a polyline and the constant offset does not affect any β-dependence.

Response. The sign convention was corrected to use $K = \nabla_S \psi \times \hat{n}$ uniformly on all four segments (commit `e70bac2`). After the fix:

- F at $\alpha = \beta = 1$, $w = 0.03$ converged to $\approx -1.8 \times 10^{-4}$ across $n_t \in \{6, 12, 18\}$, replacing the pre-fix oscillation between 5×10^{-5} and 5×10^{-4}.
- XC2 ratio improved from ~ 0.4 to ~ 0.02, comfortably passing the < 0.1 threshold.
- ℓ values stabilised to ≈ 19.03 across mesh resolutions.

The straight-wire formula was kept, with the note that this is the physically correct choice for a polyline filament.

Review of the production-sweep F data

The fits for all three α values triggered the spec's "no signal-coupled response" row of the decision table because $|f_1| < 3\sigma_{\mathrm{boot}}$. A higher-resolution self-check at $\alpha = 1, \beta \in \{0.05, 0.20\}, n_t \in \{10, 15, 20, 25\}$ showed that F does not converge with mesh refinement at this G, while ℓ remains stable to 0.02%. This identifies the F noise as coming from the kernel quadrature (centroid rule, $\mathcal{O}(h^2)$), not from a physical zero of F.

Response. The decision is reframed around the robust observable: $\ell_0 \approx 19 \gg \ell_1 \beta$ across the swept range, placing the system in the "inductance-floor kills the gain" row of the spec's decision table. This is independent of whether F is resolved, so the conclusion is unaffected by the kernel noise. An upgrade path (4-point Gauss + analytic near-singular split for adjacent panels) is noted in Sec. 8 for a future study with a G where the wire self-inductance does not dominate.